Question	Answer	Marks
2(a)	change in velocity / time (taken)	**A1**
2(b)(i)	weight ≫ (force due to) air resistance **or** (force due to) air resistance is negligible <u>compared to weight</u>	**B1**
2(b)(ii)	$s = ut + ½at^2$ $0.280 = ½ \times 9.81 \times t^2$	**C1**
	$t = 0.24$ s	**A1**

Question	Answer	Marks
2(b)(iii)	total distance fallen = 0.280 + 0.080 = 0.360 $0.360 = ½ \times 9.81 \times t^2$ $t = 0.27$ s	**C1**
	time taken = 0.27 – 0.24 = 0.03 s	**A1**
	or	
	v = 9.81 × 0.239 **or** $(2 \times 9.81 \times 0.280)^{0.5}$ **or** (2 × 0.280) / 0.239 = 2.34 ($\mathrm{m\,s^{-1}}$)	**(C1)**
	$0.080 = 2.34t + ½ \times 9.81 \times t^2$ solving quadratic equation gives $t = 0.03$ s *allow any correct method using equations of uniform accelerated motion*	**(A1)**
2(c)	(average) resultant force/acceleration/speed/velocity (of low-density ball) is less	**B1**
	(so) time interval is longer	**B1**